

PhD/Master Student/Research Assistant Progress Report

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Program: Research Assistant

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Reporting Period: 01/08/2026 - 31/08/2026

1. Progress Summary

- Focus: whether action-space curvature of latent world models explains their failure at contact-rich planning, and whether contact-aware gating of a curvature penalty repairs it. (current experiment, based on the idea of temporal straightening in latent planning)
- Outcome: the geometric pathology is real, measurable, but reducing it does not improve planning.
- Next: the residual error is not a global geometry defect, should shift to decision-critical (contact-boundary) supervision.

2. Coursework

3. Research Topic

Tentative Title: Diagnosing and Repairing Planning Misalignment in Latent World Models for Contact-Rich Manipulation

Research Problem: Latent world models are trained for predictive accuracy, while planning depends on the ordering their representations induce over candidate actions. Prediction quality therefore does not guarantee plannability, especially near contact transitions where small representation errors can flip candidate ordering and mislead CEM.

Main Objectives:

- Determine whether action-space curvature predicts and causes planner failure.
- Develop planning-aligned training using decision-critical counterfactual supervision at contact boundaries.

4. Research Progress

Key activities completed

- Built an action-space curvature diagnostic for LeWM on OGBench-Cube and validated it on a held-out snapshot set.

- Analyzed angular and radial curvature to identify which component is associated with planning failures.
- Compared three training settings: original LeWM, multi-step continuation, and continuation with an additional cosine straightening loss.
- Evaluated contact-aware gating against matched random gating.
- Performed candidate-level CEM evaluation, closed-loop planning, and same-state counterfactual analysis.

Preliminary results / outputs

- Angular curvature: higher curvature predicts more false local planning minima (+0.173, CI [+0.121, +0.235]).
- Curvature can be reduced, but the proposed cosine straightening loss did not work as expected. Multi-step continuation alone reduced false minima by 61%.
- Reducing curvature did not improve planning: CEM candidate quality showed no improvement, and closed-loop success remained unchanged (62.6% $\rightarrow$ 62.4%).
- Contact-aware gating was not better than random gating. Strong curvature regularization could also cause representation collapse instead of better physical understanding.
- A stronger remaining failure was found in candidate ranking under CEM distribution shift: ranking correlation changed from +0.247 on the initial population to -0.085 after CEM refinement.
- Same-state counterfactual analysis found deceptive candidates in 28/32 snapshots, suggesting that important errors are concentrated in small, decision-critical regions of action space.

5. Challenges and Support Needed

Challenges:

- Offline geometry and prediction gains repeatedly fail to transfer to closed-loop planning.
- Planning failure has multiple interacting sources (geometry, dynamics, ranking, search coverage), complicating causal attribution.

Support needed:

6. Plan for Next Period

Task	Deadline	Expected Output
Test best-elite execution vs elite-mean execution	10/09/2026	Determine whether the CEM execution rule is a major planning bottleneck
Re-evaluate baselines on the locked OGBench-Cube manifest	20/09/2026	Obtain results comparable with published methods

Task	Deadline	Expected Output
Finalize the next research direction	30/09/2026	Continue with operator-aware planning if positive; otherwise investigate the backbone performance gap

Student Signature: Nguyen Cuong Nhat

Supervisor Comments: